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CS607

(Artificial Intelligence)

❑ The state that represents the solution of the problem is called

- Problem statement
- Goal space
- **Solution space ✓**
- Knowledge base

❑ The Breadth-First Search traversal of a graph will result into?

- Linked List
- Arrays
- **Graph with back edges ✓**

❑ To create intelligent machines we first need to understand how the real

- introspection of mind
- psychological experiment
- **brain functions ✓**
- human body works

❑ In intelligence to solve the trivial problems we use

- hit and trial
- **trial and error ✓**
- dynamic programming
- divide and conquer

❑ The Depth First Search traversal of a graph will result in?

- Linked List

- **Tree ✓**
- Graph with back edges
- Array

❓ **Graphs are used to represent**

- terminals, branches
- **nodes, vertices ✓**
- problems, solution
- branches, nodes

❓ **Which value is assigned to alpha and beta in the alpha-beta pruning?**

- Beta = max
- Beta = min
- Alpha = max
- **Both Alpha = max & Beta = min ✓**

❓ **In Breadth-First Search, how many times a node is visited?**

- **Twice ✓**
- Thrice
- Equivalent to number of in-degree of the node

❓ **We use graphs to represent problems and their**

- nodes and vertices
- parameters
- knowledge base
- **solution spaces ✓**

❓ **"If you find the goal, exit, otherwise repeat DFS to the next lower level". The statement refers to:**

- Breadth first search
- None
- **Depth first search ✓**
- Progressive depending

1. A computer vision technique that relies on image templates is:

- ☐ robot vision
- ☒ modelbased vision
- ☐ edge detection
- ☐ pose estimation

2. The Depth First Search traversal of a graph will result in?

- ☐ Array
- ☐ Linked List
- ☒ Graph with back edges
- ☐ O O O

3. In progressive deepening, the idea is to simply apply DFS to a specific:

- ☐ Branch
- ☐ Branching factor
- ☒ Node
- ☐ O O O

4. In searches, we are concerned with finding any one solution to our problem.

- ☒ Non optimal
- ☐ Shortest path
- ☐ Optimal path
- ☐ Contrary path

5. Which of the following is true?

- ☒ A graph may contain no edges and many vertices
- ☐ A graph may contain many edges and no vertices
- ☐ A graph may contain no edges and no vertices

☐ A graph may contain no vertices and many edges

6. Which statement is wrong about heuristic search?

☐ to search and measure how far a node in a search tree seems to be from a goal

☐ to discover something or an idea embedded in a program

☒ give no optimal solution

☐ to compare two nodes in a search tree to see if one is better than another

7. General games involves:

☐ singleagent

☒ only singleagent and multiagent

☐ neither singleagent nor multiagent

☐ multiagent

8. How many types of graphs are there to use in problemsolving?

☒ 0 1

☐ 0

9. Graphs are used to represent:

☐ nodes, vertices

☐ branches, nodes

☐ terminals, branches

☒ problems, solution

10. In BreadthFirst Search, how many times is a node visited?

☐ Thrice

☒ Once

☐ Twice

O Equivalent to number of indegree of the node

1. Which of the following is not a branch and bound strategy to generate branches?

Options:

- A. LIFO branch and bound
- B. Lowest cost branch and bound
- C. Highest cost branch and bound
- D. FIFO branch and bound

Correct Option: B. Lowest cost branch and bound

2. To create intelligent machines we first need to understand how the real _____.

Options:

- A. brain functions
- B. human body works
- C. human body works
- D. introspection of mind

Correct Option: A. brain functions

3. "In context of Hill climbing algorithm, you might just reach local maxima and think that you have reached the global maxima, so getting stuck in the middle of searching the solution space." The above statement refers to:

Options:

- A. Foothill problem
- B. Ridge problem
- C. Plateau problem
- D. Plateau problem

Correct Option: C. Plateau problem

4. A natural language generation program must decide _____.

Options:

- A. what to say**
- B. both what to say & when to say something**
- C. why it is being used**
- D. when to say something**

Correct Option: B. both what to say & when to say something

5. In intelligence to solve the trivial problems we use _____ approach.

Options:

- A. trial and error**
- B. divide and conquer**
- C. hit and trial**
- D. dynamic programming**

Correct Option: A. trial and error

6. Which of the following is not the application area of intelligence?

Options:

- A. Expert system**
- B. Robotics**
- C. Computer vision**
- D. Human computer interaction**

Correct Option: D. Human computer interaction

7. A "circle" symbol in a tree structure is called:

Options:

- A. Edge**
- B. Descendant**
- C. Node**
- D. Ancestor**

Correct Option: C. Node

8. Which approach is used by the Best First Search algorithm while searching?

Options:

A. Greedy Combinatorial

B. Heuristic

C. Divide and Conquer

D. None of the above

Correct Option: A. Greedy Combinatorial

9. The Data structure used in the standard implementation of DepthFirst Search is?

Options:

A. Stack

B. recursion

C. Queue

D. Linked List

Correct Option: A. Stack

10. There are many techniques to solve our problem of optimal search without using a brute force technique; one such procedure is called _____.

Options:

A. Progressive deepening

B. Breadth first method

C. Depth first method

D. Branchandbound method

Correct Option: D. Branchandbound method

1. A search algorithm takes _____ as an input and returns _____ as an output.

Options:

- a) Solution, problem
- b) Problem, solution
- c) Fuzzy set, uncertain facts
- d) Input, output

Correct option: d) Input, output

2. In intelligence to solve the trivial problems we use _____ approach.

Options:

- a) Dynamic programming
- b) Hit and trial
- c) Trial and error
- d) Hit and trial

Correct option: d) Hit and trial

3. One of the advantages of breadthfirst search is that it:

Options:

- a) Has a small space requirement
- b) Can work with broken edges
- c) Memory Constraints
- d) Guarantees finding the shallowest path even in presence of infinite paths

Correct option: d) Guarantees finding the shallowest path even in presence of infinite paths

4. "In context of Hill climbing algorithm, situation may arise when a step in one direction takes you lower, on the other hand when you step in some other direction it gives you no improvement." The above statement refers to:

Options:

- a) Plateau problem
- b) Ridge problem
- c) Slope problem

d) Foothill problem

Correct option: a) Plateau problem

5. The simple idea behind _____ is that if we can reach a specific node through more than one different paths then we shall take the path with the minimum cost.

Options:

a) Beam search

b) Dynamic programming

c) Estimates

d) Progressive deepening

Correct option: b) Dynamic programming

6. Where do the values of the alphabeta search get updated?

Options:

a) At the end

b) Along the path of search

c) At the beginning

d) Initial state itself

Correct option: b) Along the path of search

7. In the minmax procedure, the player hoping for the positive numbers is called _____.

Options:

a) Researcher minimizer

b) Analyzer

c) Minimizer

d) Maximizer

Correct option: d) Maximizer

8. Which search is equal to minmax search but eliminates the branches that can't influence the final decision?

Options:

- a) Alphabeta pruning
- b) Beam search
- c) Breadthfirst search
- d) Depthfirst search

Correct option: a) Alphabeta pruning

9. A "circle" symbol in a tree structure is called:

Options:

- a) Node
- b) Edge
- c) Ancestor
- d) Descendant

Correct option: a) Node

10. If an arrow points from node "A" to Node "B" then, node "B" will be called:

Options:

- a) Child of node "A"
- b) Parent of node "A"
- c) Fanout of node "A"
- d) Branching factor of node "A"

Correct option: a) Child of node "A"

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